# BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, PESHAWAR

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2022

# MATHEMATICS

Past Paper 2022 • Paper Code: Mathematics Paper–X (01) (22)

Retyped faithfully from the original board paper — same questions, no changes.

## SECTION-A

**Time Allowed:** 20 Minutes | **Maximum Marks:** 15 | **Paper Code:** Mathematics Paper–X (01) (22)

۱۔ ہر سوال کے سامنے چار دائرے دیے گئے ہیں۔ صرف صحیح جواب والا دائرہ بھر دیں۔ ۲۔ دائروں کو شیڈ (بھرنے) کے لیے نیلے یا کالے رنگ کا پین استعمال کریں۔ ۳۔ جواب میں ایک سے زائد دائرے بھرنے سے جواب غلط تصور ہوگا۔

Q1. In.....circles congruent chords are equidistant from the centers.

(A) Semi
(B) Equal
(C) Congruent
(D) Great

Q2. A line that intersects the circle at two distinct points is called.....to a circle.

(A) Tangent
(B) Secant
(C) Normal
(D) Chord

Q3. The measure of the entire circle is.....

(A) 360°
(B) 180°
(C) 90°
(D) 45°

Q4. The angle in a segment less than a semi circle is.....

(A) Acute
(B) Right
(C) Obtuse
(D) 45°

Q5. Product of the roots of $mx^2 - 5x + n = 0$ is.....

(A) $-n/m$
(B) $-5/m$
(C) $m/n$
(D) $n/m$

Q6. If A={1,3,5,7} and B={3,4,5,6,7,8} then A \ B = .....

(A) {1}
(B) {8}
(C) {2}
(D) {}

Q7. If 'A' and 'B' are any two disjoint sets then A ∩ B = .....

(A) {}
(B) {0}
(C) [illegible in source scan]
(D) [illegible in source scan]

Q8. The difference of largest and smallest values in the data is called.....

(A) Mean
(B) Range
(C) Frequency
(D) Interval

Q9. $1 + \cot^2\theta$ = .....

(A) $\tan^2\theta$
(B) $\sec^2\theta$
(C) $\operatorname{cosec}^2\theta$
(D) $-\operatorname{cosec}^2\theta$

Q10. Number of radians in one complete rotation is .....radians.

(A) π/4
(B) π/2
(C) π
(D) 2π

Q11. If $x^2 - 1/4 = 0$ then the solution are x =.....

(A) ±16
(B) ±1/16
(C) ±1/2
(D) ±1/4

Q12. The roots of the equation $ax^2 + bx + c = 0$ are unequal and imaginary if

(A) $b^2 - 4ac < 0$
(B) $b^2 - 4ac = 0$
(C) $b^2 - 4ac > 0$
(D) $b^2 - 4ac \leq 0$

Q13. In simplified form 6a : 18b is.....

(A) a : b
(B) a : 3b
(C) 3a : b
(D) 6a = 18b

Q14. If 7 : 9 :: x : 27 then x = .....

(A) 21
(B) 63
(C) 81
(D) 229

Q15. Resolution of $x/(x+1)^2$ into partial fractions = .....

(A) $A/(x+1) + B/(x+1)$
(B) $A/(x+1) + B/(x+2)$
(C) $A/(x+1) + B/(x+1)^2$
(D) $(Ax+b)/(x+1)^2$

## Section-B

**Time Allowed:** 2 Hours 40 Minutes | **Maximum Marks:** 36

> Note: Note: Time allowed for Section – B and Section – C is 2 Hours and 40 minutes.

QII. Attempt any NINE parts. Each part carries FOUR marks.

1. Solve by completing square method $x^2 - 8x + 9 = 0$
2. Solve the equation $\sqrt{x} + \sqrt{(3x+1)} = \sqrt{(5x+1)}$
3. Find the cube roots of 8.
4. If $\alpha, \beta$ are the roots of $2x^2 + 3x + 1 = 0$, then find the value of $\alpha/\beta + \beta/\alpha$.
5. If $3x - 1, 4, 35$ are in continued proportion, then find the value of x.
6. Volume of gas 'V' varies inversely as pressure 'P'. $P = 300\ N/m^2$ when $V = 4m^3$. Find pressure when $V = 3m^3$
7. Resolve into partial fractions $(2x-1)/(x(x-3)^2)$
8. Domain of binary relation $R = \{(x,y) / y + 1 = 2x^2\}$ is set N. Find its range.
9. If $U = \{1,2,3,4,5,6,7\}$, $A = \{3,5,6\}$, $B = \{2,3,4\}$ then draw Venn Diagrams for A' and B'.
10. Find the standard deviation for the data 6,8,10,12,14
11. Prove that $\sin\theta \cdot \sqrt{(1+\tan^2\theta)} = \tan\theta$
12. Find the area of sector of a circle whose radius is 4m, with central angle 12 radians.

## Section-C

**Time Allowed:** 2 Hours 40 Minutes | **Maximum Marks:** 24

> Note: Note: Time allowed for Section – B and Section – C is 2 Hours and 40 minutes.

Q. Attempt any THREE questions. All questions carry equal marks.

QIII. Perpendicular from the centre of a circle on a chord bisects it. Prove it.

QIV. Prove that if two circles touch externally, the distance between their centers is equal to the sum o their radii.

QV. Prove that equal chords of a circle subtend equal angles at the centre of the circle. Prove for one circle.

QVI. Circumscribe a square about a circle of radius 5 cm.